

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12403011

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Orphanos; Stephen J. et al.

Baseplate of a modular shoulder joint prosthesis and related methods for implanting the same

Abstract

Shoulder joint implants are disclosed herein for use in shoulder reconstruction that are configured to facilitate the inclusion a central bone screw for augmented bone fixation. The implant can include a baseplate (or metaglene) configured to secure a glenosphere or other prosthetic component to the glenoid. To facilitate lateral or proximal insertion of a central bone screw through the implant, a throughbore defined along the central axis of the metaglene can be widened to accommodate the maximum diameter of the screw. To enable fixation of the glenosphere to the metaglene, a collet can be configured to engage the smaller diameter of a glenosphere coupling element and inserted into the central throughbore. The collet and the bone screw can be separate parts, thereby making insertion of the bone screw optional. Alternatively, the collet and bone screw can be integrated together to form a unitary construct.

Inventors: Orphanos; Stephen J. (Bridgewater, MA), Otrando; Brian (Cumberland, RI), Murphy; William (Brockton, MA), Reppucci; Steven (Lakeville, MA)

Applicant: DePuy Ireland Unlimited Company (Ringaskiddy, IE)

Family ID: 67438751

Assignee: DePuy Ireland Unlimited Company (Ringaskiddy, IE)

Appl. No.: 18/444376

Filed: February 16, 2024

Prior Publication Data

Document Identifier

Publication Date

US 20240341967 A1

Oct. 17, 2024

Related U.S. Application Data

Publication Classification

Int. Cl.: A61F2/40 (20060101)

U.S. Cl.:

CPC A61F2/4014 (20130101); A61F2/4081 (20130101); A61F2002/4018 (20130101)

Field of Classification Search

CPC: A61F (2/40); A61F (2/014); A61F (2/4081); A61F (2002/4018)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4045825	12/1976	Stroot	N/A	N/A
4045826	12/1976	Stroot	N/A	N/A
5314479	12/1993	Rockwood, Jr. et al.	N/A	N/A
5358526	12/1993	Tornier	N/A	N/A
5376126	12/1993	Lin	N/A	N/A
5728161	12/1997	Camino et al.	N/A	N/A
5782920	12/1997	Colleran	N/A	N/A
5876459	12/1998	Powell	N/A	N/A
5906644	12/1998	Powell	N/A	N/A
5961555	12/1998	Huebner	N/A	N/A
6005018	12/1998	Cicierega et al.	N/A	N/A
6033439	12/1999	Camino et al.	N/A	N/A
6102953	12/1999	Huebner	N/A	N/A
6129764	12/1999	Servidio	N/A	N/A
6168627	12/2000	Huebner	N/A	N/A
6168628	12/2000	Huebner	N/A	N/A
6171342	12/2000	O	N/A	N/A
6193758	12/2000	Huebner	N/A	N/A
6508840	12/2002	Rockwood, Jr. et al.	N/A	N/A
6589281	12/2002	Hyde, Jr.	N/A	N/A
6592622	12/2002	Ferguson	N/A	N/A
6620197	12/2002	Maroney et al.	N/A	N/A
6723129	12/2003	Dwyer et al.	N/A	N/A
6736851	12/2003	Maroney et al.	N/A	N/A
6783549	12/2003	Stone et al.	N/A	N/A
6863690	12/2004	Ball et al.	N/A	N/A
6953478	12/2004	Bouttens et al.	N/A	N/A

6969406	12/2004	Tornier	N/A	N/A
6984248	12/2005	Hyde, Jr.	N/A	N/A
6986790	12/2005	Ball et al.	N/A	N/A
7011686	12/2005	Ball et al.	N/A	N/A
7044973	12/2005	Rockwood, Jr. et al.	N/A	N/A
7169184	12/2006	Dalla Pria	N/A	N/A
7175664	12/2006	Lakin	N/A	N/A
7198642	12/2006	Hazebrouck et al.	N/A	N/A
7297163	12/2006	Huebner	N/A	N/A
7338498	12/2007	Long et al.	N/A	N/A
7527631	12/2008	Maroney et al.	N/A	N/A
7608109	12/2008	Dalla Pria	N/A	N/A
7611539	12/2008	Bouttens et al.	N/A	N/A
7615080	12/2008	Ondrla	N/A	N/A
7753959	12/2009	Berelsman et al.	N/A	N/A
7854768	12/2009	Wiley et al.	N/A	N/A
7918892	12/2010	Huebner	N/A	N/A
8002838	12/2010	Klotz	N/A	N/A
8070755	12/2010	Maroney et al.	N/A	N/A
8070820	12/2010	Winslow et al.	N/A	N/A
8118868	12/2011	May et al.	N/A	N/A
8182542	12/2011	Ferko	N/A	N/A
8192497	12/2011	Ondrla	N/A	N/A
8273093	12/2011	Klotz et al.	N/A	N/A
8303665	12/2011	Tornier et al.	N/A	N/A
8419798	12/2012	Ondria et al.	N/A	N/A
8425614	12/2012	Winslow et al.	N/A	N/A
8460390	12/2012	Biss et al.	N/A	N/A
8469999	12/2012	Gonzalez-Hernandez	N/A	N/A
8579984	12/2012	Borowsky	N/A	N/A
8591592	12/2012	Dreyfuss	N/A	N/A
8632597	12/2013	Lappin	N/A	N/A
8689425	12/2013	Mutchler et al.	N/A	N/A
8702762	12/2013	Jacene et al.	N/A	N/A
8814943	12/2013	Long et al.	N/A	N/A
8920508	12/2013	Iannotti et al.	N/A	N/A
8940054	12/2014	Wiley et al.	N/A	N/A
8945138	12/2014	Klotz et al.	N/A	N/A
8968410	12/2014	Veronesi et al.	N/A	N/A
8979940	12/2014	Porter et al.	N/A	N/A
9044330	12/2014	Chavarria et al.	N/A	N/A
9084680	12/2014	Katrana et al.	N/A	N/A
9107758	12/2014	Long et al.	N/A	N/A
9283075	12/2015	Wiley et al.	N/A	N/A
9345580	12/2015	Porter et al.	N/A	N/A
9351743	12/2015	Kehres et al.	N/A	N/A
9421105	12/2015	Metcalfe et al.	N/A	N/A
9421106	12/2015	Splieth et al.	N/A	N/A

9433508	12/2015	Phipps	N/A	N/A
9445911	12/2015	Long et al.	N/A	N/A
9498344	12/2015	Hodorek et al.	N/A	N/A
9597190	12/2016	Chavarria et al.	N/A	N/A
9610165	12/2016	Poncet et al.	N/A	N/A
9629724	12/2016	Lappin	N/A	N/A
9629725	12/2016	Gargac et al.	N/A	N/A
9668873	12/2016	Winslow et al.	N/A	N/A
9763798	12/2016	Chavarria et al.	N/A	N/A
9770334	12/2016	Wiley et al.	N/A	N/A
10064734	12/2017	Burkhead, Jr. et al.	N/A	N/A
10251744	12/2018	Treacy et al.	N/A	N/A
10383735	12/2018	Wiley et al.	N/A	N/A
10390972	12/2018	Rao	N/A	N/A
10799335	12/2019	Treacy et al.	N/A	N/A
10813759	12/2019	Drasler et al.	N/A	N/A
10813769	12/2019	Orphanos et al.	N/A	N/A
11432935	12/2021	Paterson et al.	N/A	N/A
11458019	12/2021	Cleveland et al.	N/A	N/A
11903839	12/2023	Orphanos	N/A	A61F 2/4014
2002/0004685	12/2001	White	N/A	N/A
2003/0139818	12/2002	Rogers et al.	N/A	N/A
2003/0149485	12/2002	Tornier	N/A	N/A
2004/0193278	12/2003	Maroney et al.	N/A	N/A
2004/0220673	12/2003	Pria	N/A	N/A
2005/0004679	12/2004	Sederholm et al.	N/A	N/A
2005/0033443	12/2004	Blatter et al.	N/A	N/A
2005/0251263	12/2004	Forrer et al.	N/A	N/A
2006/0142872	12/2005	Klotz et al.	N/A	N/A
2007/0100458	12/2006	Dalla Pria	N/A	N/A
2007/0112430	12/2006	Simmen et al.	N/A	N/A
2007/0288020	12/2006	Yang et al.	N/A	N/A
2008/0208348	12/2007	Fitz	N/A	N/A
2008/0269894	12/2007	Melvin	N/A	N/A
2008/0281428	12/2007	Meyers et al.	N/A	N/A
2008/0294268	12/2007	Baum et al.	N/A	N/A
2009/0171462	12/2008	Poncet et al.	N/A	N/A
2009/0192621	12/2008	Winslow et al.	N/A	N/A
2009/0254188	12/2008	Maroney et al.	N/A	N/A
2010/0161066	12/2009	Iannotti et al.	N/A	N/A
2011/0009973	12/2010	Meyers et al.	N/A	N/A
2011/0054625	12/2010	Ferko et al.	N/A	N/A
2011/0106267	12/2010	Grant	N/A	N/A
2011/0130840	12/2010	Oskouei	N/A	N/A
2011/0224799	12/2010	Stone	N/A	N/A
2012/0239051	12/2011	de Wilde et al.	N/A	N/A
2012/0253467	12/2011	Frankle	N/A	N/A
2012/0277880	12/2011	Winslow et al.	N/A	N/A
2013/0030540	12/2012	Leibinger	N/A	N/A
2013/0066433	12/2012	Veronesi et al.	N/A	N/A

2013/0090736	12/2012	Katrana et al.	N/A	N/A
2013/0150972	12/2012	Iannotti et al.	N/A	N/A
2013/0150973	12/2012	Splieth et al.	N/A	N/A
2013/0261751	12/2012	Lappin	N/A	N/A
2013/0338780	12/2012	Berchoux et al.	N/A	N/A
2015/0305877	12/2014	Gargac et al.	N/A	N/A
2015/0335440	12/2014	Linares et al.	N/A	N/A
2016/0008138	12/2015	Katrana et al.	N/A	N/A
2016/0030180	12/2015	Wecker et al.	N/A	N/A
2016/0100835	12/2015	Linder et al.	N/A	N/A
2016/0270922	12/2015	Pressacco et al.	N/A	N/A
2017/0042690	12/2016	Burkhead, Jr. et al.	N/A	N/A
2017/0079803	12/2016	Lang	N/A	N/A
2017/0100251	12/2016	Ek et al.	N/A	N/A
2017/0156847	12/2016	Ricci et al.	N/A	N/A
2017/0172751	12/2016	Poncet et al.	N/A	N/A
2017/0181860	12/2016	Nerot et al.	N/A	N/A
2017/0273800	12/2016	Emerick et al.	N/A	N/A
2017/0354508	12/2016	Chavarria et al.	N/A	N/A
2018/0021150	12/2017	Harper et al.	N/A	N/A
2018/0078377	12/2017	Gargac et al.	N/A	N/A
2018/0161169	12/2017	Cardon et al.	N/A	N/A
2018/0214261	12/2017	Treacy et al.	N/A	N/A
2018/0303551	12/2017	Hopkins	N/A	N/A
2019/0046326	12/2018	Ball	N/A	N/A
2020/0030108	12/2019	Orphanos et al.	N/A	N/A
2020/0085563	12/2019	John	N/A	N/A
2020/0155202	12/2019	Jackson	N/A	A61B 17/864
2020/0405491	12/2019	Cleveland et al.	N/A	N/A
2021/0007857	12/2020	Orphanos et al.	N/A	N/A
2021/0128201	12/2020	Samuel et al.	N/A	N/A
2021/0251768	12/2020	Nelson	N/A	N/A
2022/0008103	12/2021	McPhee	N/A	A61B 17/8605
2022/0008206	12/2021	LaReau	N/A	N/A
2022/0026543	12/2021	Bolatkale et al.	N/A	N/A
2022/0362025	12/2021	Cleveland et al.	N/A	N/A
2022/0362030	12/2021	Patkar et al.	N/A	N/A
2023/0000526	12/2022	Follini et al.	N/A	N/A
2023/0000636	12/2022	Dalla Pria et al.	N/A	N/A
2023/0265952	12/2022	Ansell	285/242	F16L 33/224
2024/0115295	12/2023	Klausman	N/A	A61B 17/8605

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H09276305	12/1996	JP	N/A
2013086440	12/2012	WO	N/A
2017007565	12/2016	WO	N/A

OTHER PUBLICATIONS

[No Author Listed] Delta Xtend, Reverse Shoulder System—Surgical Technique, DePuy Synthes, 2017 (66 pages). cited by applicant
Extended European Search Report for EP Application No. 19188150.7 issued Oct. 23, 2019 (8 pages). cited by applicant
India Examination Report for Application No. 201914027853, issued Sep. 22, 2022 (6 pages). cited by applicant
Japanese Notice of Reason for Refusal for Application No. 2019-135327, dated Sep. 5, 2023 (13 pages). cited by applicant
U.S. Appl. No. 16/044,473, filed Jul. 24, 2018, Baseplate of a Modular Shoulder Joint Prosthesis and Related Methods for Implanting the Same. cited by applicant
U.S. Appl. No. 17/036,706, filed Sep. 29, 2020, Baseplate of a Modular Shoulder Joint Prosthesis and Related Methods for Implanting the Same. cited by applicant
Chinese Office Action for Application No. 201910671069 dated Aug. 5, 2023 (18 pages). cited by applicant

Primary Examiner: Stewart; Alvin J

Attorney, Agent or Firm: Barnes & Thornburg LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of and claims priority to U.S. patent application Ser. No. 17/036,706, filed Sep. 29, 2020, which is a continuation of and claims priority to U.S. patent application Ser. No. 16/044,473, filed Jul. 24, 2018, entitled “Baseplate of a Modular Shoulder Joint Prosthesis and Related Methods for Implanting the Same,” and now issued as U.S. Pat. No. 10,813,769, the content of each which is hereby incorporated by reference herein in its entirety.

FIELD

(1) The present disclosure relates generally to orthopedic implants, and more particularly to a baseplate of a modular shoulder joint prosthesis and related methods of implanting the baseplate into a patient's scapula.

BACKGROUND

(2) During the lifetime of a patient, it may be necessary to perform a total shoulder replacement procedure on the patient as a result of disease or trauma, for example. In a total shoulder replacement procedure, a humeral prosthesis is used to replace the natural head of the patient's humerus. The humeral prosthesis typically includes an elongated post component that is implanted into the intramedullary canal of the patient's humerus and a hemispherically-shaped prosthetic head component that is secured to the post component. In such a total shoulder replacement procedure, the natural glenoid surface of the scapula is resurfaced or otherwise replaced with a glenoid component that provides a bearing surface upon which the prosthetic head component of the humeral prosthesis articulates.

(3) However, in some cases the patient's natural shoulder, including its soft tissue, has degenerated to a severe degree of joint instability and pain. In many such cases, it can be necessary to change the mechanics of the shoulder. Reverse shoulder implants can be used to do so. As its name suggests, a reverse shoulder implant reverses the anatomy, or structure, of the healthy shoulder. In particular, a reverse shoulder implant is designed such that the prosthetic head (i.e., the “ball” in the ball-and-socket joint), known as a glenosphere component, is secured to the patient's scapula, with

the corresponding concave bearing (i.e., the “socket” in the ball-and-socket joint), known as a humeral cup, being secured to the patient's humerus. Such a reverse configuration allows the patient's deltoid muscle, which is one of the larger and stronger shoulder muscles, to raise the arm.

(4) To secure the glenosphere component to the patient's scapula, a baseplate, sometimes referred to as a metaglene component, can be implanted onto the glenoid of the patient's scapula. For example, as shown in FIGS. 1A and 1B, a shoulder joint implant includes a metaglene component **60** that includes a platform **62** having a post **64** extending outwardly from its distal surface **66**. The post **64** has a bore **68** formed about a central axis A-A of the metaglene component **60** and is configured to engage a locking screw or other coupling element protruding from a distal surface of a glenosphere component (not shown). The bore **68** extends through the entire length of the post **64**. The post **64** of the metaglene component **60** is designed to be implanted into a void formed in a glenoid surface **28**. Bone screws **80** can be positioned in some or all of the screw holes or apertures **74** and driven into, or otherwise secured to, the bone tissue of a patient's scapula **30**, thereby fixing the metaglene component **60** in place.

(5) In some challenging clinical situations, such as poor bone quality or anatomic limitations, it can be useful to drive a bone screw into the glenoid through the central bore **68** of the metaglene component **60**. However, when the proximal width or diameter of the bore **68** is smaller than the width or diameter of the head of the screw, the surgeon may be prevented from inserting the screw through the bore using a top-down or lateral approach (i.e., distally inserting the screw from the proximal end of the metaglene component), thereby increasing the complexity of the surgical procedure. Although the proximal diameter of the bore **68** can be widened to accommodate the screw diameter, such widening can cause backward incompatibility with glenoid/glenosphere components having coupling elements with narrower widths.

(6) Accordingly, there is a need for improved metaglene components that, for example, facilitate use of a central screw, and related methods for implanting the various embodiments of a metaglene component provided for herein, or otherwise derivable from the present disclosures, in a scapula of a patient with (or without) a central screw in a manner that maintains backward compatibility with standard glenoid/glenosphere components.

SUMMARY

(7) The present disclosure is generally related to shoulder joint implants, and represents improvements over existing designs of such implants. These implants are often used in reverse shoulder procedures. Typically the shoulder joint implant includes a baseplate, sometimes referred to herein as a metaglene component (or metaglene). The base plate that is configured to secure a glenosphere component (or glenosphere) to the bony anatomy of the glenoid to complete the repair. As discussed in greater detail below, other components that can engage with the baseplate and/or the glenosphere component can include a collet and/or a bone screw, and in some such instances the collet and bone screw can form a singular component of the implant. The provided for embodiments can allow for backwards compatibility amongst the components of the implant, and also allow for surgeons to select a variety of different configuration types for the implant based, at least in part, on the needs and anatomy of the patient and the preferences of the particular surgeon. For example, some configurations that include a collet can allow a bone screw to be omitted from the implant, while some other configurations that include a collet and bone screw coupled together in a single, unitary component can reduce the number of steps performed during a procedure, thereby simplifying the procedure.

(8) In one exemplary embodiment of a shoulder joint implant, the implant includes a first prosthetic component, a baseplate, and a removable collet. The first prosthetic component includes a coupling element that protrudes from a distal bearing surface of the prosthetic component. The baseplate is configured to secure the first prosthetic component to bone, and includes a post that protrudes from a distal bearing surface of the baseplate. The post defines a throughbore that extends along a proximal-distal axis of the post. The removable collet is disposed within the throughbore, and is

configured to engage the coupling element of the first prosthetic component such that the first prosthetic component is fixedly coupled to the baseplate.

(9) In some embodiments, the first prosthetic component is a glenosphere component having an integrated locking screw protruding from a distal bearing surface of the glenosphere component. The collet can be dimensioned to engage the coupling element, with the coupling element having a diameter that is less than a diameter of the throughbore of the baseplate. Additionally, or alternatively, the collet can include a plurality of proximally-extending arms that are radially compressible to facilitate at least one of insertion and removal of the collet in the throughbore of the baseplate. One or more of the proximally-extending arms can include a lateral protrusion that is configured to engage an annular recessed portion of an inner sidewall of the throughbore when the arms are not compressed, which in turn can fix the removable collet in place with respect to the first prosthetic component and/or with respect to the baseplate itself at a predetermined depth in the throughbore of the baseplate. Further, one or more of the proximally-extending arms can have a proximal ramped bearing surface that is configured to bear against a counterpart bearing surface of a prosthetic implant tool. The arm(s) can be radially compressed in response to the counterpart bearing surface of the prosthetic implant tool bearing against the proximal ramped bearing surface, thus disengaging the lateral protrusion of the arm(s) from the recessed portion of the inner sidewall of the throughbore. This radial compression feature can be used to facilitate insertion and/or removal of the collet within the throughbore of the baseplate.

(10) In some embodiments, the collet can include one or more keys protruding from one or more of the arms of the collet. The keys of the collet can be configured to engage with one or more keyways defined longitudinally along an inner sidewall of the throughbore of the baseplate to align and guide the collet into the baseplate. The keys of the collet can be configured to engage with the keyways defined along the inner sidewall of the throughbore to resist rotation of the collet while disposed within the throughbore of the baseplate. The keys of the collet can be configured to engage with the keyways along the inner sidewall of the throughbore to form a locking mechanism that provides a torsional resistive force that counteracts a torque imparted by the coupling element of the first prosthetic component while screwing the coupling element into the collet to form a taper lock between the first prosthetic component and the baseplate or while unscrewing the coupling element out of the collet to break the taper lock between the first prosthetic component and the baseplate. The length of the keyways can define a distal most depth at which the removable collet can be seated in the throughbore of the baseplate. In some embodiments, the distance between the annular recessed portion and the distal most depth of the keyways can be a predefined distance that is greater than the height of the collet such that the collet bottoms out at the distal end of the one or more keyways while the coupling element of the glenosphere component is unscrewed from the collet.

(11) In some embodiments, the implant can include a bone screw that has a proximal head and a distal threaded portion. In such embodiments, the throughbore of the baseplate can be dimensioned to facilitate insertion of the bone screw that has a diameter that is greater than a diameter of the coupling element of the first prosthetic component. Further, in some such embodiments, the collet can be dimensioned to engage the coupling element, with the coupling element having a diameter that is less than a diameter of the central bone screw. In some instances, the collet and the central bone screw can be separate components. This can result, for example, in a configuration in which the collet is disposed within the throughbore of the baseplate, proximal to the proximal head of the central bone screw. In some such embodiments, the collet can be disposed within the throughbore of the baseplate after the central bone screw is laterally inserted into the throughbore. In some embodiments, the implant can be devoid of a central bone screw.

(12) In instances in which the implant includes a central bone screw that has a proximal head and a distal threaded portion, the collet can be coupled to the proximal head of the central bone screw. Alternatively, or additionally, the proximal head of the central bone screw can include a shoulder

portion that is of a spherical and/or conical shape, and an inner sidewall of the throughbore at the distal end of the post can include a threaded sidewall portion that is configured to support the shoulder portion of the central bone screw. In some embodiments that include a central bone screw that has a proximal head and a distal threaded portion, the distal threaded portion of the central bone screw can extend through an opening at the distal end of the post, being configured to secure the baseplate to bone.

(13) One exemplary method of implanting a shoulder prosthetic implant in a scapula of a patient includes reaming a glenoid of a scapula such that a surface of the glenoid is complementary to a distal bearing surface of a baseplate of a prosthetic shoulder implant, drilling a hole into the glenoid to receive a post that protrudes from the distal bearing surface of the baseplate, and disposing the post of the baseplate into the glenoid through the drilled hole. The method further includes securing a plurality of peripheral bone screws into the glenoid through one or more screw holes defined in the baseplate, and laterally inserting a removable collet into a throughbore defined in the baseplate. The throughbore extends along a proximal-distal axis of the baseplate of the post. Still further, the method includes securing a first prosthetic component of the prosthetic shoulder implant onto a proximal bearing surface of the baseplate such that a coupling element protruding from a distal bearing surface of the first prosthetic component is secured to the collet within the throughbore of the baseplate.

(14) The method can further include laterally inserting a central bone screw through the throughbore of the baseplate prior to inserting the removable collet, and securing a distal threaded portion of the central bone screw into the glenoid through an opening at the distal end of the post of the baseplate. The collet can be coupled to a proximal head of a central bone screw and the method can further include laterally inserting the collet and the central bone screw together into the throughbore of the baseplate, and securing a distal threaded portion of the central bone screw into the glenoid through an opening at the distal end of the post of the baseplate.

(15) The throughbore of the baseplate can be dimensioned to facilitate insertion of a central bone screw having a diameter that is greater than a diameter of the coupling element of the first prosthetic component. Further, the collet can be dimensioned to engage the coupling element of the first prosthetic component having a diameter that is less than a diameter of the central bone screw. In some embodiments, the first prosthetic component is a glenosphere component having an integrated locking screw protruding from a distal bearing surface of the glenosphere component.

(16) The removable collet can include a plurality of compressible arms. In some such embodiments, the method can include removing the collet from within the throughbore of the baseplate using a tool by laterally inserting the tool into the throughbore of the baseplate, and rotating a distal end of the tool to engage the threaded portion of the plurality of arms. The method can further include continuing to rotate the distal head until the distal head radially compresses the plurality of arms away from a recessed sidewall portion of the throughbore, and proximally pulling the collet out of the throughbore.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate exemplary embodiments, and together with the general description given above and the detailed description given below, serve to explain the features of the various embodiments:

(2) FIG. 1A is a side, partial cross-sectional view of one example of a shoulder joint implant of the prior art;

(3) FIG. 1B is a schematic side view of the shoulder joint implant of FIG. 1A coupled to a scapula;

- (4) FIG. 2A is a perspective view of one exemplary embodiment of a shoulder joint implant;
- (5) FIG. 2B is a perspective exploded view of components of the shoulder joint implant of FIG. 2A, the components including a metaglene component, a collet, a glenosphere component, and a bone screw;
- (6) FIG. 3 is a side, partially transparent view of the shoulder joint implant of FIG. 2B;
- (7) FIG. 4A is a bottom perspective view of the metaglene component of FIG. 2B;
- (8) FIG. 4B is a detailed top view of the metaglene component of FIG. 4A;
- (9) FIG. 4C is a detailed side, cross-sectional view of the metaglene component of FIG. 4B;
- (10) FIG. 5 is a perspective view of the collet of FIG. 2B;
- (11) FIG. 6A is a side perspective view of the bone screw of FIG. 2B;
- (12) FIG. 6B is a detailed perspective view of a distal end the bone screw of FIG. 6A;
- (13) FIG. 6C is a detailed side, cross-sectional view of the metaglene component of FIG. 4C and a proximal end of the bone screw of FIG. 6A;
- (14) FIG. 7 is a detailed side, cross-sectional view of the metaglene component and the proximal end of the bone screw of FIG. 6A and the collet of FIG. 5;
- (15) FIG. 8 is a side, cross-sectional view of the metaglene component, proximal end of the bone screw, and the collet of FIG. 7, and the glenosphere component of FIG. 2B that includes a locking screw;
- (16) FIG. 9 is a schematic illustration of one exemplary embodiment of a method of implanting a shoulder implant like the shoulder implant of FIGS. 2A-8;
- (17) FIG. 10A is a perspective exploded view of another exemplary embodiment of a shoulder joint implant, the implant having components that include a metaglene component, a glenosphere component, and an integrated collet and bone screw;
- (18) FIG. 10B is a side, partially transparent view of the shoulder joint implant of FIG. 10A;
- (19) FIG. 10C is a detailed perspective view of the integrated collet and bone screw of FIG. 10A;
- (20) FIG. 11 is a schematic illustration of one exemplary embodiment of a method of implanting a shoulder implant like the shoulder implant of FIGS. 10A-10C;
- (21) FIG. 12A is a perspective view of one exemplary embodiment of a collet removal tool;
- (22) FIG. 12B is a side perspective, cross-sectional view of one exemplary embodiment of the collet removal tool of FIG. 12A being associated with a shoulder joint implant that includes a collet;
- (23) FIG. 12C is a detailed side, partially transparent view of the collet removal tool and collet of FIG. 12B;
- (24) FIG. 13A is a perspective view of yet another exemplary embodiment of a shoulder joint implant;
- (25) FIG. 13B is a perspective exploded view of components of the shoulder joint implant of FIG. 13A, the components including a metaglene component having an integrated bone screw and a glenosphere component;
- (26) FIG. 14 is a side, partially transparent view of the shoulder joint implant of FIG. 13B; and
- (27) FIG. 15 is a schematic illustration of one exemplary embodiment of a method of implanting a shoulder joint implant like the shoulder implant of FIGS. 13A and 13B.

DETAILED DESCRIPTION

(28) Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present disclosure is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of

the present disclosure. Sizes and shapes of the devices, and the components thereof, can depend on a variety of factors, including but not limited to an anatomy and tendencies of the subject (i.e., patient) in which the devices will be used, the size and shape of components with which the devices will be used, the methods and procedures in which the devices will be used, and the preferences of the surgeon operating the devices and/or otherwise performing the related procedure(s).

(29) In the present disclosure, like-named components of the embodiments generally have similar features and/or purposes, unless stated otherwise. Additionally, a number of terms may be used throughout the disclosure interchangeably but will be understood by a person skilled in the art. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed devices and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such devices and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can be easily determined for any geometric shape (e.g., references to widths and diameters being easily adaptable for circular and linear dimensions, respectively, by a person skilled in the art). Additionally, to the extent that terms are used in the disclosure to describe a direction, orientation, and/or relative position of the disclosed prosthetic devices and components thereof and/or for performing a disclosed method of assembly and/or implantation of such devices, such terms are not intended to be limiting. For example, a person skilled in the art will recognize that terms of direction, orientation, and/or relative position (e.g., proximal, distal, medial, lateral, etc.) can be used interchangeably depending, at least in part, on the perspective view of the surgeon or other operator.

(30) The present disclosure is generally related to shoulder joint implants, and represents improvements over existing designs of such implants. These implants are often used in reverse shoulder procedures. Typically the shoulder joint implant includes a baseplate, sometimes referred to herein as a metaglene component (or metaglene). The baseplate is configured to secure a glenosphere component (or glenosphere) to the bony anatomy of the glenoid to complete the repair. As discussed in greater detail below, other components that can engage with the metaglene component and/or the glenosphere component can include a collet and/or a bone screw, and in some such instances the collet and bone screw can form a singular component of the implant. The provided for embodiments can allow for backwards compatibility amongst the components of the implant, and also allow for surgeons to select a variety of different configuration types for the implant based, at least in part, on the needs and anatomy of the patient and the preferences of the particular surgeon. For example, some configurations that include a collet can allow a bone screw to be omitted from the implant, while some other configurations that include a collet and bone screw coupled together in a single, unitary component can reduce the number of steps performed during a procedure, thereby simplifying the procedure.

(31) Although the various embodiments disclosed herein disclose fixing a glenosphere component to the metaglene component, one of skill in the art will recognize the other prosthetic components can be coupled to the metaglene component, such as but not limited to a glenoid component configured to replace or augment the glenoid surface of the scapula. A person skilled in the art will also understand how the disclosures provided for herein can be adapted for use with devices and procedures associated with other joints without departing from the spirit of the present disclosure.

(32) FIGS. 2A, 2B, and 3 provide for one exemplary embodiment of a shoulder joint implant **200**. As shown, the shoulder joint implant **200** can include a metaglene (or metaglene component) **210**, a collet **230**, a glenosphere (or glenosphere component) **250**, and a central bone screw **270**, which, as explained below, can be optional. More particularly, the collet **230** and the bone screw **270** can be disposed in a central throughbore of the metaglene **210** such that the bone screw can be driven into the glenoid and the collet can secure the glenosphere **250** to the metaglene **210**. Each of these components is described in greater detail below.

(33) The metaglene **210** is one component of the shoulder joint implant **200**, and is illustrated in more detail in FIGS. 4A-4C. In the illustrated embodiment, the metaglene **210** includes a platform

312 having a post **314** extending outwardly from a bottom or distal-facing surface **312d**. The platform **312** can be generally configured such that its bottom or distal-facing surface **312d** engages in a complementary manner with a surface of the glenoid when implanted and a top or proximal-facing surface **312p** engages in a complementary manner with the glenosphere **250** or other prosthetic component when coupled together. More specifically, the distal-facing surface **312d** of the platform **312** can have a convex shape configured to conform to a concave surface of the glenoid. Further, the proximal-facing surface **312p** of the platform **312** can have a concave shape configured to be complementary with a surface it engages with of the glenosphere **250**.

(34) The platform **312** can include a central throughbore **310** extending through a thickness of the platform **312**. Its configuration is described in greater detail below with respect to its relationship with the post **314** and the collet **230**. Although illustrated in a central configuration, a person skilled in the art will recognize the throughbore **310** can be located in a non-central manner without departing from the spirit of the present disclosure. The platform **312** can also include one or more screw holes or apertures **318** that likewise extend through a thickness of the platform **312**. The apertures **318** can be positioned in any variety of configurations, and in the illustrated embodiment there are four that are positioned radially outward from the central throughbore **310** the such that they are approximately equidistant from each other. The apertures **318** can be used to receive peripheral bone screws (not shown, but operate in a similar manner as screws **80** of FIGS. **1A** and **1B**) to be driven into the bony anatomy of the glenoid to help fix the metaglene **210** in place. Any number and configuration of apertures **318** can be used without departing from the spirit of the disclosure, including using no apertures.

(35) The post **314** can generally extend distally from the distal-facing surface **312d** of the platform **312**, and can be configured to be press fit into a void or hole drilled into the glenoid of the patient's scapula, thereby providing a first form of bone fixation. In some embodiments, to further augment bone fixation, the post **314** can have a throughbore **320** extending therethrough. In the illustrated embodiment the throughbore **320** is centrally disposed such that it is formed and centered around a central axis A-A of the metaglene **210**, although other locations with respect to the metaglene **210** are possible. This location also places the throughbore **320** in-line with the throughbore **310** of the platform **312**. The throughbore **320** can have an inner diameter sized to accommodate a maximum diameter of a bone screw **270** (e.g., the diameter of a screw head **372**). For example, as shown in the illustrated embodiment, the bone screw **270** can be inserted laterally or proximally through the throughbore **320** of the metaglene **210** such that the screw head **372** is seated on a threaded distal portion **316** of the post **314**. The threaded distal portion **374** of the screw **270** exits the metaglene **210** through a distal opening **308** of the post **314**. As described in more detail below with respect to FIG. **7**, the threaded distal portion **316** of the post **314** can be configured to facilitate the use of smaller screws and thus avoid the need to increase the size of the post **314**. A distal opening **308** of the post **314** can be used to allow a central bone screw (not shown) to exit the metaglene **210** into bone.

(36) As shown in FIGS. **4B** and **4C**, the central throughbore **310** of the metaglene **210** can be configured to facilitate insertion of the collet **230** and optionally a bone screw **270**. For example, in some embodiments, one or more slots or recesses, sometimes referred to herein as keyway(s) **322**, can be defined longitudinally along an inner sidewall **310s** of the throughbore **310**. The keyways can be configured to slidably mate with keys **512** protruding from the collet **230**, described in greater detail below, to help orient and guide insertion of the collet **230** within the throughbore **310** of the metaglene **210**. The keyways **322** can extend partially along the length of the throughbore **310** and terminate at a distal stop **322d** at which to seat the collet **230**.

(37) In some embodiments, the mating of the keys **512** and keyways **322** can provide resistance to torsional rotation, e.g., while the glenosphere **250** is screwed into or otherwise coupled to the collet **230**. For example, in some embodiments, the keys **512** of the collet **230** can be configured to engage with the keyways **322** along the inner sidewall **310s** of the throughbore **310** to form a

locking mechanism that provides a torsional resistive force that counteracts a torque imparted by the locking screw of the glenosphere **250** while screwing the locking screw into the collet **230** to form a taper lock (e.g., a Morse taper lock) between the distal facing surface of the glenosphere **250** and the proximal facing surface of the baseplate **210**. Likewise, the mating of the keys **512** and the keyways **322** form a locking mechanism that provides a torsional resistive force that counteracts a torque imparted by the locking screw in an opposite direction while unscrewing the coupling element out of the collet to break the taper lock between the first prosthetic component and the baseplate. As shown in the illustrated embodiment, the keyways **322** can be shaped to form rectangular or other suitably shaped slots. Although two keyways are shown in the illustrated embodiment, more or less than two keyways (e.g., four) can be defined within the throughbore **310** of the metaglene **210**.

(38) Alternatively, or additionally, in some embodiments the central throughbore **310** of the metaglene **210** can be configured to define an annular groove or recess, sometimes referred to herein as a catch **324**, in the inner sidewall **310s** of the central throughbore **310**. As described in more detail with respect to FIG. 7, the annular catch **324** can be configured to engage the lip **514** of the collet, thereby securing the collet **230** in place at a predetermined depth.

(39) Alternatively, or additionally, in some embodiments a threaded distal portion **316** can be defined along the inner sidewall **310s** of the central throughbore **310** at the distal end of the post **314**. In some embodiments, the threaded distal portion **316** of the central throughbore **310** can include two or more threads **316t** configured to coarsely match the threads of a central bone screw (e.g., **270**). As shown in FIG. 4C, the threads **316t** of the distal threaded portion **316** can be configured at an oblique angle or otherwise radially contoured to serve as a spherical or conical bearing surface for the head of the central bone screw. As described in more detail with respect to FIG. 6A-6C, the distal threaded portion **316** of the throughbore **310** can allow the use of a central bone screw having a maximum thread size without having a large screw head, and thereby avoiding the need to increase the size of the post.

(40) The platform **312** and post **314** can come in a variety of configurations, shapes, and sizes. As shown, the base plate **312** can be substantially cylindrical with a substantially circular cross-section. As discussed above, a top surface **312p** of the platform **312** can have a concave configuration, and a bottom surface **312d** of the platform can have a convex configuration. The post **314** can also be substantially cylindrical with a substantially circular cross-section, with a diameter of the post **314** being less than a diameter of the platform **312**. In the illustrated configuration, the diameter of the post **314** is approximately half the size of the diameter of the platform **312**, although other proportions are certainly possible, such as the diameter of the post **314** being approximately one-quarters or three-quarters the size of the diameter of the platform **312**.

(41) While the sizes of the components can certainly vary, in some exemplary embodiments a diameter of the platform **312** can be approximately in the range of about 15 millimeters to about 30 millimeters, a thickness of the platform **312** (as viewed, proximal-to-distal) can be approximately in the range of about 3.5 millimeters to about 14 millimeters, and a radius of curvature of the concavity of either or both the proximal-facing and distal-facing surfaces **312p**, **312d** of the platform **312** being approximately in the range of about 0.5 millimeters to about 2.0 millimeters. A person skilled in the art will recognize that a thickness of the platform **312** may change over its surface area of the radius of curvature of the concavity of the platform for its proximal-facing surface **312p** is not the same as the radius of curvature of the concavity of the platform for its distal-facing surface **312d**. A diameter of the post **314** can be approximately in the range of about 5 millimeters to about 20 millimeters, and a thickness of the post **314** (as viewed, proximal-to-distal) can be approximately in the range of about 5 millimeters to about 30 millimeters. In some exemplary embodiments, a diameter of the platform **312** is approximately 22 millimeters or approximately 27 millimeters, a thickness of the platform **312** is approximately 7 millimeters,

approximately 8 millimeters, or approximately 10 millimeters, a radius of curvature of the concavity of each of the proximal-facing surface **312p** and the distal-facing surface **312d** of the platform **312** is approximately 1.0 millimeters, a diameter of the post **314** is approximately 10 millimeters, and a thickness of the post can be approximately 13 millimeters, approximately 23 millimeters, or approximately 28 millimeters.

(42) The glenosphere **250** is illustrated in FIGS. 2A, 2B, and 3. In some embodiments, the glenosphere **250** can include a hemispherical body **352** having a locking screw **354** or other coupling element that extends outward from a distal surface **356** of the hemispherical body **352**. The distal surface **356** can be complementary in size and shape to the proximal-facing surface **312p** of the metaglene **210** so that they can form a secure fit when coupled together. In the illustrated embodiment, the fit is a thread fit, but in other embodiments it could be a snap fit or any other coupling technique known to those skilled in the art for coupling two mechanical components. While many different sizes, shapes, and configurations of the glenosphere **250** are possible, in some exemplary embodiments a diameter of the hemispherical body **352** is approximately in the range of about 20 millimeters to about 80 millimeters, such as 40 millimeters, and a height of the hemispherical body **352** (as viewed, proximal-to-distal) can be approximately in the range of about 10 millimeters to about 40 millimeters, such as 20 millimeters. Likewise, in some exemplary embodiments, a diameter of the locking screw **354** can be approximately in the range of about 2.5 millimeters to about 10 millimeters, such as 5 millimeters, and a height of the locking screw **354** (as viewed, proximal-to-distal) can be approximately in the range of about 12.5 millimeters to about 50 millimeters, such as 25 millimeters.

(43) In previous iterations of shoulder joint implants, a glenosphere locking screw was screwed directly into a central throughbore of a metaglene to secure the glenosphere to the metaglene. However, when the diameter of the central throughbore of the metaglene is widened to accommodate the diameter of the head of a bone screw, the diameter of the central throughbore can exceed the diameter or width of a glenosphere locking screw such that the locking screw cannot be screwed directly to the metaglene.

(44) To address this problem, the present disclosure provides for the collet **230**, which is illustrated in FIG. 5 and sometimes referred to as an adaptor or a reducing bushing. As shown in the illustrated embodiment, the collet **230** can include a substantially hollow cylindrical body **502**. The collet can also include open-ended slots **504** spaced about the circumference of the cylindrical body **502** to form radially compressible arms **506** that extend proximally from an annular base portion **508**. As described in more detail below with respect to FIG. 7, the radially compressible arms of the collet **230** can be configured to facilitate insertion and/or removal of the collet by flexing in response to changes in the diameter of the throughbore **310** of the metaglene **210**. In some embodiments, each of the proximally-extending arms **506** can have an inner threaded portion **510** configured to at least partially surround and threadably engage the locking screw **354** of the glenosphere **250** or other prosthetic component.

(45) In some embodiments, the collet **230** can include one or more lateral protrusions that are configured to orient and guide the collet into the throughbore of the metaglene, sometimes referred to herein as key(s) **512**. For example, as shown in the illustrated embodiment, the keys **512** can have a rectangular or other suitable shaped profile and protrude radially outward from one or more of the arms **506** and the annular base portion **508** of the collet **230**. As described in more detail below with respect to FIG. 7, the keys **512** can be configured to slide along the keyways **322** formed along an inner sidewall of the throughbore **310** of the metaglene **210**. When mated to the inner sidewall of the throughbore, the keys **512** can provide resistance to torsional rotation, e.g., while the glenosphere **250** is screwed into or otherwise coupled to the collet **230**.

(46) In some embodiments, the collet **230** can include an annular protrusion, sometimes referred to herein as a ridge or lip **514**. As described in more detail below with respect to FIG. 7, the lip can be disposed or segmented across one or more of the arms **506** and configured to engage a catch **324**

(e.g., an annular groove or recess) defined within the throughbore of the metaglene, thereby securing the collet **230** in place at a predetermined depth.

(47) In some embodiments, the collet **230** can include a tapered or conical proximal bearing surface **516** at the proximal end of each arm **506**. As described in more detail below with respect to FIGS. **12A-12C**, the conical or tapered bearing surface **516** can be used to facilitate removal of the collet from within the central throughbore of the metaglene.

(48) The collet **230** can be disposed within the central throughbore **310** to effectively reduce the inner diameter of the throughbore and threadably engage or otherwise capture the locking screw **354** of the glenosphere **250**. In some embodiments, the collet **230** can have an inner threaded portion **332** that enables the glenosphere locking screw **354** to be screwed directly into the collet and thereby draw the glenosphere **250** onto the proximal-facing bearing surface **312p** of the metaglene **210**.

(49) The collet **230** can be laterally or proximally inserted into the throughbore **310** of metaglene **210** until the collet **230** reaches a predetermined depth within the throughbore **310**. In other words, the collet **230** is advanced distally towards and into the throughbore **310**. As previously described, in some embodiments, the keyways **322** and/or the catch **324** defined in the sidewall **310s** of the central throughbore **310** can be used to engage the keys **514** and/or the lip **514** of the collet **230** and thereby position the collet at a predetermined depth. Alternatively, or additionally, the collet **230** can be inserted into the throughbore of the metaglene component after the central bone screw **270** is laterally or proximally inserted into the throughbore, such the collet **230** can be disposed proximal to a head of the central bone screw **270**. In some embodiments the collet **230** can be inserted into the throughbore of the metaglene component with no central bone screw **270** disposed therein.

(50) Like the other components, the collet **230** can have a variety of sizes, shapes, and configurations. In some exemplary embodiments, the collet has a diameter approximately in the range of about 3 millimeters to about 15 millimeters, such as about 6.4 millimeters, and a height (as viewed, proximal-to-distal) approximately in the range of about 5 millimeters to about 20 millimeters, such as about 10 millimeters. Although shown as a cylindrical shape, other shapes are possible provided that the shape is complementary to the shape of the throughbore **310** of the metaglene **210** into which it is inserted.

(51) FIGS. **6A**, **6B**, and **6C** illustrate a central bone screw **270** that is configured for use with the shoulder joint implant of FIG. **2A**. As shown in the illustrated embodiment, the central screw **270** can have a screw head **602**, a shoulder portion **604**, an unthreaded neck portion **606**, a distal threaded portion **608** and a distal tip **610**. In some embodiments, the shoulder portion **604** can be shaped to form a spherical or conical bearing surface **604d** for the screw head **602**. Notably, the central bone screw **270** is an optional component that can be omitted while still allowing the rest of the shoulder joint implant **200** to be successfully used for a repair procedure.

(52) Typical screw heads are generally much larger in diameter than the thread diameter in order to provide a sufficient bearing surface. However, in some embodiments, in view of the present disclosures, the screw head **602** can have an outer diameter that is the same or approximately the same as the maximum outer diameter of the distal threaded portion **608**. In such embodiments, the spherical or conical shoulder portion **604** can serve as the bearing surface for the screw head. For example, the shoulder portion **605** of the screw can provide a spherical or conical bearing surface **604d** that extends inward at an oblique angle between the head portion **602** and the neck portion **606**.

(53) Thus, as shown in the illustrated embodiment of FIG. **6C**, when the screw **270** is inserted through the central throughbore **310** and seated at the distal opening **308** of the metaglene post **314**, the spherical or conical bearing surface **604d** of the screw can bear against the distal threaded portion **316** of the throughbore **310**. For example, as shown, the threads **316t** of the distal threaded portion **316** can be angled or otherwise radially contoured to match or approximately match the

spherical or conical bearing surface **604d** of the screw **270**. Thus, the distal threaded portion **316** can serve as a proximal-facing bearing surface that substantially conforms to shaped profile of the shoulder portion **604** to support the screw head **602** within the post **314**. As shown in the illustrated embodiment, the spherical or conical shoulder portion **604** can be unthreaded to prevent it from passing through the distal threaded portion **316** of the baseplate, such that the shoulder portion can serve as the bearing surface of the screw **604** to the baseplate **210**. Accordingly, in some embodiments the distal threaded portion **316** of the central throughbore **310** can allow the use of a central bone screw in which the screw head and threaded portion are very close in size (e.g., diameter) and still provide a bearing surface to provide optimal compression. Additionally, the use of central bone screws having reduced diameter heads can provide an advantage in that increases in the size (e.g., outer diameter) of the metaglene post can be avoided and thus requiring removal of less bone in the glenoid.

(54) In some embodiments, the unthreaded neck portion **606** of the bone screw **270** can be cylindrical or substantially cylindrical in shape. The unthreaded neck portion **606** can be configured to provide a clearance between the spherical or conical shoulder portion **604** to the threaded portion **608**. The amount of such clearance provided by the neck portion **606** can be configured so that the distal threaded portion **608** of the screw **270** does not mate directly to the metaglene **210** when the screw head **602** is seated within the post **314**.

(55) In some embodiments, the unthreaded neck portion **606** of the screw **606** can allow a “free spinning screw” design in which the screw is disposed within the baseplate, as shown in FIG. 3. The unthreaded neck portion **606** can help prevent the distal threaded portion **608** from binding up within the post **314** of the baseplate **210**. Accordingly, the unthreaded neck portion **606** allows the screw **600** to turn within the bone and thereby drag the baseplate **210** further into the bone to provide additional compression between the bone and the distal facing surface **312d** of the platform **312**. This feature allows the screw to act like any other socket head cap screw that is assembled into a counter-bored hole to provide compression.

(56) In some embodiments, the unthreaded neck portion **606** can provide an additional bearing surface or contact surface area between the screw **600** and baseplate **210**. Additionally, the unthreaded neck portion **606** can be spherical which allows the screw **600** to pivot off axis from the baseplate **210** and thus not influence the baseplate **210** to glenoid bone interface. In some embodiments, the unthreaded neck portion **606** can be omitted such that the screw **606** could still function as intended with less contact area.

(57) In some embodiments, the diameter and length of the screw **270** can be varied to accommodate user needs. For example, in some embodiments, the length of the screw **270** can be approximately in the range of about 15 millimeters to about 80 millimeters, and in some embodiments it is approximately 40 millimeters. However, in some embodiments, the surgeon can be provided with a number of central bone screws having different screw lengths from which to choose depending, at least in part, on the depth of the glenoid vault in which the screw is to be secured. In some embodiments, the maximum outer diameter of the distal threaded portion **608** of the screw **270** can be approximately in the range of about 6.0 millimeters to about 6.5 millimeters. In some embodiments, the distal threaded portion **640** of the screw **270** can be configured for optimized purchase or grip within the glenoid vault and allow for single or multiple leads to aid in insertion speed. In some embodiments, the distal tip **610** of the screw **270** can include an optimized bone cutting self-tapping feature **612**. The self-tapping feature **612** as shown is a channel formed in an outer surface of the distal threaded portion **640**, which makes it easier to insert the screw **270** into bone.

(58) In view of the present design, and the fact that in at least some instances a bone screw may not be needed, the surgeon can choose to not use the bone screw **270** in the assembly of the shoulder joint implant **200**. If the central bone screw **270** is not used, the collet **230** can be inserted in the metaglene **210** without the bone screw. In some embodiments, the shoulder joint implant **200** can

be fixedly attached to the bony anatomy of the glenoid as part of a reverse shoulder reconstruction procedure, described in more detail with respect to FIG. 9.

(59) FIG. 7 illustrates the collet **230** and bone screw **270** disposed within the central throughbore **310** of the metaglene **210**. In the illustrated embodiment, the collet **230** and the bone screw **270** are separate components. Thus, the surgeon can choose whether or not to include the central bone screw **270** in the assembly of the shoulder joint implant **200**. For example, in some embodiments, if the surgeon chooses to include the central bone screw in the assembly, the screw **270** can be laterally or proximally inserted through the central throughbore **310** (i.e., advanced distally towards and into the metaglene **210**). Thereafter, the collet **230** can be laterally or proximally inserted into the central throughbore **310** such the collet **230** is disposed proximal to the head **602** of the screw **270**. Alternatively, if the surgeon chooses not to include the central bone screw **270**, the collet **230** can be laterally or proximally inserted into the central throughbore **310** without no bone screw.

(60) A detailed description of the structure and procedure for laterally or proximally inserting and seating a central bone screw **270** within the central throughbore **310** is described above with respect to FIGS. 6A-6C and thus is omitted here for the sake of brevity.

(61) In some embodiments, the collet **230** can be laterally or proximally inserted into the central throughbore **310** of the metaglene **210** by aligning the keys **512** of collet **230** with the keyways **322** of the throughbore **310**. For example, as previously discussed with respect to FIGS. 4A-4C and 5, the collet **230** can include one or more lateral protrusions or keys **512** that are configured to slidably mate with corresponding slots or keyways **322** defined in the throughbore **310**. Once the collet **230** is oriented by aligning the keys **512** with the keyways **322**, the collet **230** can be pushed laterally or proximally into the central throughbore **310** such that it is guided by the keys **512** sliding laterally or proximally along the respectively keyways **322**.

(62) In some embodiments, as the collet **230** enters the throughbore **310**, the inner sidewall **310s** of throughbore **310** bears against the ridge or lip **514** of collet **230** and thereby causes the radially compressible arms **506** of the collet to flex inward while remaining in contact with the wall. As previously discussed with respect to FIGS. 4A-4C and 5, the lip **514** of the collet **230** can be an annular protrusion disposed across one or more of the arms **506** and configured to engage an annular groove or catch **324** defined horizontally along the inner sidewall **310s** of the throughbore **310**.

(63) In some embodiments, when the lip **514** of the collet **230** reaches the location of the catch **324**, the radial compression force applied by the inner sidewall **310s** can release and cause the collet arms **506** to flex outward such that the lip snaps into or otherwise mates within the catch. In some embodiments, the engagement of the lip **514** of the collet **230** and the catch **324** can create an audible and/or tactile cue (e.g., a “click”) to notify the surgeon that the collet has reached the desired depth. In some embodiments, the engagement of the lip **514** of the collet **230** with the catch **324** can coincide with the keys **512** at the annular base portion **508** of the collet **230** hitting a mechanical stop **322d** at the bottom of the keyways **322**. With the collet **230** secured at the predetermined depth, a glenosphere (e.g., the glenosphere **250**) or other prosthetic component can be attached to the metaglene **210** using techniques described herein or otherwise known to those skilled in the art.

(64) In some embodiments, the location of the mechanical stop **322d** at the distal end of the keyways **322** can define a distal most depth, or the key depth, at which the distal end of the collet **230** is capable of bottoming out within the throughbore **310** of the baseplate **210**. The distance **D** between the distal most depth of the keyways **322** and the annular recessed portion defined in the inner sidewall **310s** of the throughbore **310**, or catch **324**, can be a predefined distance that is slightly greater than the height of the collet **230**. For example, in some embodiments, the distance **D** between the distal most depth of the keyways and the catch **324** can be greater than the height of the collet by an amount in the range between approximately 0.5 millimeters and approximately 2.0 millimeters, such as about 1.0 millimeters. In some embodiments, where the height of the collet

230 is in the range between approximately 5 millimeters and 20 millimeters, the distance **D** can be in the range between approximately 4.0 millimeters and 22 millimeters. For example, in some embodiments, where the height of the collet **230** is approximately 10 millimeters, the distance **D** can be in the range between 9 and 12 millimeters, such as about 11 millimeters. Thus, the additional tolerance provided by the predefined distance **D** between the catch **324** and the distal end **322d** of the keyways **322** can allow the collet **230** to be properly secured between the catch **324** and the distal end **322d** of the keyways **322**.

(65) The additional tolerance provided by the predefined distance **D** can be useful during removal of the glenosphere (e.g., **250**). For example, in some embodiments, the glenosphere and thus the glenosphere locking screw (e.g., **354**) can be unscrewed (e.g., rotated counterclockwise) to remove the glenosphere from the baseplate **210**. As the locking screw is unscrewed, the ridge or lip **514** of the collet **230** releases from the catch **324** in the sidewall **310s** of the throughbore **310**, thereby causing the collet to be pushed towards the mechanical stop **322d** of the keyways **322** until the distal end of the collet **230** bottoms out against the stop. Once the collet **230** bottoms out against the mechanical stop **322d**, the continuing turning of the locking screw **354** to unscrew the glenosphere (e.g. **250**) causes the load to be transferred through glenosphere such that the taper lock connection between the glenosphere and the baseplate **210** is pulled apart. In some embodiments, the taper lock connection can be a Morse taper lock between the distal facing surface of the glenosphere and the proximal facing surface of the baseplate **210**. This is essentially a “Jacking Screw” function for removal of the glenosphere from the baseplate.

(66) FIG. **8** illustrates the shoulder joint implant **200** with the glenosphere **250** secured to the collet **230**. As shown in the illustrated embodiment, the collet **230** can be disposed within the central throughbore **310** to effectively reduce the inner diameter of the throughbore **310** and threadably engage or otherwise capture the locking screw **354** of the glenosphere **250**. For example, as shown, each of the proximally-extending collet arms **506** can have an inner threaded portion **510** configured to at least partially surround the locking screw **354** such that a threaded portion **358** of the locking screw **354** can be screwed into the collet **230** and thereby draw and secure the hemispherical body **352** of the glenosphere **250** onto the proximal-facing surface **312** of the metaglene **210**. When the collet **230** is secured to the inner sidewall **310s** of the throughbore **310**, the engagement of the keys **512** and the keyways **322** can provide resistance to torsional rotation, e.g., while the glenosphere **250** is screwed into or otherwise coupled to the collet **230**. Alternatively or in addition, engagement of the lip **514** of the collet **230** and the catch **324** can prevent or otherwise limit vertical displacement of the collet **230** within the throughbore **310**.

(67) In exemplary embodiments any and all of the metaglene **210**, the collet **230**, the glenosphere **250**, and the screw **270** can be made from any number of implantable metallic materials or other biocompatible materials to form the implant. Some non-limiting examples of materials suitable for forming the various components implant can include titanium, cobalt-chrome, stainless steel, and other metals known to those skilled in the art, and some plastic materials, such as but not limited to, polyetheretherketone (PEEK) and Ultra High Molecular Weight Polyethylene (UHMWPE). In some instances, the various components (e.g., the metaglene **210**, the collet **230**, the glenosphere **250**, and the screw **270**) can be made from the same material, while in other embodiments one or more components can be made from different materials. For example, in some instances in may be beneficial to have the metaglene **210**, the collet **230**, and the glenosphere **250** formed from surgical grade titanium, while the screw **270** is formed from surgical grade titanium coated with a porous biocompatible material (e.g., a hydroxyapatite (HA) coating). Additionally, a person skilled in the art will recognize that a number of different material blends can be used to form any component of the implants provided for herein or otherwise derivable from the present disclosures.

(68) FIG. **9** illustrates an exemplary embodiment of a method **900** of implanting the shoulder joint implant **200** in a patient's scapula. In the illustrated embodiment, the shoulder joint implant **200** can be implanted as part of a reverse shoulder reconstruction surgery in which the glenosphere (i.e., the

“ball” in the ball-and-socket joint) is secured to the patient's scapula. A humeral cup (i.e., the “socket” in the ball-and-socket joint) can be secured to the patient's humerus to provide a corresponding concave bearing for the glenosphere. Such a reverse configuration allows the patient's deltoid muscle, which is one of the larger and stronger shoulder muscles, to raise the arm. Although the illustrated configuration uses the implant **200** described above, at least some of the components associated with or otherwise used in conjunction with the implant **200** may not be easily visible in the various blocks **910**, **920**, **930**, **940**, **950**, **960**, and **970**. In view of the disclosures provided for herein, and their related illustrations, a person skilled in the art will understand how the various components of the implant **200** engage the various portions of the patient's anatomy, and/or the components of the implant **200** and related tools used in conjunction with the procedures disclosed with respect to FIG. 9.

(69) At block **910**, a glenoid **912** of the patient's scapula **914** can be reamed such that a bony surface of the glenoid **912** conforms to the distal bearing surface (e.g., **312d**) of the metaglene (e.g., **210**). In some embodiments, a glenoid resurfacing reamer **710** can be used to prepare the glenoid **912** to have an approximately smooth, curved surface having the same or substantially the same radius of curvature as the distal-facing surface **312d** of the metaglene **210**. The diameters may also be complementary, whether that means the same or substantially equal, or the glenoid **912** being a little larger with the labrum (not visible in block **910**) being adjacent to help keep the metaglene **210** in the desired location.

(70) At block **920**, a hole can be drilled into the glenoid **912** to receive the metaglene post (e.g., **314**) of the metaglene (e.g., **210**). In some embodiments, a cannulated stop drill **720** aligned with a central guide pin (not shown) can be used to drill the central hole.

(71) At block **930**, the metaglene post **314** (not visible) can be disposed into the glenoid **912** through the drilled hole. In some embodiments, a metaglene delivery instrument **730** can be secured to the metaglene **210** and used to insert the metaglene post **314** into the hole drilled into the glenoid **912**. The metaglene post **314** can be slightly oversized relative to the diameter of the drilled hole to enable a press fit. Once the post **314** and the distal-facing surface **312d** (not visible) of the metaglene **210** is seated on the bone **912**, the metaglene delivery instrument **730** can be removed.

(72) At block **940**, one or more peripheral bone screws (not shown) can be driven into the glenoid through one or more screw apertures **318** of the metaglene **210** to secure the metaglene **210** to bone. In some embodiments, a screw driver **740** can be used to screw the peripheral bone screws at superior and/or inferior locations in the glenoid **912**. In some embodiments, peripheral bone screws can be placed at lateral and/or medial locations of the glenoid.

(73) At optional block **950**, a bone screw **270** (not visible) can be laterally or proximally inserted through the central throughbore **310** (not visible) of the metaglene **210** prior to inserting the collet **230**. In some embodiments, a screw driver **750** can be used to screw a distal threaded portion **608** (not visible) of the bone screw **270** into the glenoid through an opening **308** (not visible) at the distal end of the post **314** (not visible) of the metaglene **210**. In some embodiments, the bone screw can be laterally or proximally inserted as previously described with respect to FIGS. 6A-6C.

(74) At block **960**, the collet **230** (not visible) can be laterally or proximally inserted into the central throughbore **310** (not visible) defined in the metaglene **210** into the post **314** (not visible). In some embodiments, a collet delivery instrument **760** can be coupled to the collet **230** (not visible) and used to push the collet **230** into the central throughbore **310** (not visible) as previously described with respect to FIG. 7.

(75) At block **970**, the glenosphere **250** of the shoulder joint implant **200** can be secured onto the proximal bearing surface **312p** of the metaglene **210**. For example, in some embodiments, a locking screw **354** (not visible) or other coupling element protruding from a distal bearing surface of the glenosphere **250** can be secured to the collet **230** (not visible) within the throughbore of the metaglene **210**. In some embodiments, the glenosphere **250** can be screwed into or otherwise coupled the collet **230** as previously described with respect to FIG. 8.

(76) FIGS. 10A and 10B provide for another exemplary embodiment of a shoulder joint implant **1000**. As shown, the shoulder joint implant **1000** can include a metaglene **1010**, a glenosphere **1030**, and an integrated collet and central bone screw **1050**. Accordingly, the integrated collet and central bone screw **1050** can allow surgeons to add a bone screw down the center axis of the metaglene component as a single unitary component in one step, while maintaining backwards compatibility with existing glenosphere components.

(77) Except as described below or as will be readily appreciated by one having ordinary skill in the art, the metaglene **1010** and the glenosphere **1030** are substantially similar to the metaglene **210** and the glenosphere **250** described above with respect to FIGS. 2A-9. A detailed description of the structure and function thereof is omitted here for the sake of brevity. The shoulder joint implant **1000** can include any one or more of the features of the shoulder joint implant **200**.

(78) As shown in the illustrated embodiment, the integrated collet and bone screw **1050** can include a collet portion **1052**, a proximal screw head **1054**, a spherical or conical shoulder portion **1056**, an unthreaded neck portion **1058**, a distal threaded portion **1060**, and a distal tip **1062**. Except as described below or as will be readily appreciated by one having ordinary skill in the art, the shoulder portion **1056**, an unthreaded neck portion **1058**, a distal threaded portion **1060**, and a distal tip **610** are substantially similar to the spherical or conical shoulder portion **604**, the unthreaded neck portion **606**, the distal threaded portion **608**, and the distal tip **610** of the screw **270** described above with respect to FIGS. 6A-6C. A detailed description of the structure and function thereof is omitted here for the sake of brevity.

(79) In some embodiments, the collet portion **1052** can be fixedly coupled to the proximal head **1054** of the screw to form the integrated collet and central bone screw **1050**. For example, the collet portion **1052** can be mechanically attached, welded, bonded, or attached in any other manner of ways to the proximal screw head **1054**. Alternatively, the collet portion **1052** can be integrally formed on the proximal screw head **1054**, e.g., using a three dimensional (3D) printing technique.

(80) As shown in the illustrated embodiment of FIG. 10C, the collet portion **1052** can have a substantially hollow cylindrical body **1070**. The collet portion **1052** can include open-ended slots **1072** spaced about the circumference of the cylindrical body **1070** to form radially compressible arms **1074** that extend proximally from an annular base portion **1076**. The radially compressible arms **1074** of the collet portion **1052** can be configured to facilitate insertion and/or removal of the collet by flexing in response to changes in the diameter of the throughbore of the metaglene **1010**.

(81) In some embodiments, each of the proximally-extending arms **1074** can have an inner threaded portion **1078** configured to at least partially surround and threadably engage a locking screw **1032** of the glenosphere **1030** or other prosthetic component. In some embodiments, the collet portion **1052** can include an annular protrusion, sometimes referred to herein as a lip **1080**. As shown, the lip **1080** can be disposed or segmented across one or more of the arms **1074** and configured to engage an annular groove or recess defined within the throughbore of the metaglene (e.g., catch **324** of FIG. 4C), thereby securing the collet **230** in place at a predetermined depth. In some embodiments, the collet portion **1052** can include a tapered or conical proximal bearing surface **1082** at the proximal end of each arm **1074**. As described in more detail below with respect to FIGS. 12A and 12B, the conical or tapered bearing surface **1082** can be used to facilitate removal of the collet.

(82) In some embodiments, the integrated collet/central bone screw **1050** can be laterally or proximally inserted into the central throughbore of the metaglene **1010** and driven into the bony anatomy of the glenoid in as single unitary component. For example, in some embodiments, the integrated collet and central bone screw **1050** can be advanced into the central throughbore of the metaglene (e.g., **310** of FIG. 4C) until the distal tip **1062** of the screw **1050** reaches the threaded distal portion (e.g., **316** of FIG. 4C) of the central throughbore. Once the integrated collet/central bone screw **1050** reaches the threaded distal portion **316**, the integrated collet/central bone screw can be screwed loosely through the threaded distal portion to advance the unitary component into

bony anatomy of the glenoid.

(83) In some embodiments, as the collet portion **1052** of the unitary component enters the throughbore, the inner sidewall of the throughbore (e.g., **310s** of FIG. **4C**) bears against the lip **1080** of collet portion **1052** and thereby causes the radially compressible arms **1074** of the collet to flex inward while remaining in contact with the wall. The lip **1080** of the collet portion **1052** can be an annular protrusion disposed across one or more of the arms **1074** and configured to engage an annular groove or catch (e.g., **324** of FIG. **4C**) defined horizontally along the inner sidewall **310s** of the throughbore **310**.

(84) In some embodiments, when the lip **1080** of the collet portion **1052** reaches the location of the catch **324**, the radial compression force applied by the inner sidewall **310s** can release and cause the collet arms **1074** to flex outward such that the lip snaps into or otherwise mates within the catch. In some embodiments, the engagement of the lip **1080** of the collet portion **1052** and the catch **324** can create an audible and/or tactile cue (e.g., a “click”) to notify the surgeon that the collet has reached the desired depth. With the collet portion **1052** secured at the predetermined depth, a glenosphere **1030** or other prosthetic component can be screw or otherwise attached to the metaglene **1010**.

(85) FIG. **11** illustrates an exemplary embodiment of a method **1100** of implanting the shoulder joint implant **1100** in a patient's scapula. In the illustrated embodiment, the shoulder joint implant **1000** can be implanted as part of a reverse shoulder reconstruction surgery. The method **1100** may include operations in blocks **910**, **920**, **930**, **940**, and **970** that are described above with reference to FIG. **9**. Accordingly, a detailed description of these operations is omitted here for the sake of brevity.

(86) At block **1110**, the integrated collet and central bone screw **1050** (not visible) can be laterally or proximally inserted into the central throughbore defined in the metaglene **1010** through the post and driven into the glenoid through an opening at the distal end of the post of the metaglene **1010**. In some embodiments, the integrated collet-bone screw **1050** can be coupled to the distal head of a delivery tool **770**. The tool **770** can be operated by the surgeon to push the integrated collet-bone screw **1050** into the central throughbore (e.g., **310**, not visible) and then to rotate integrated collet-bone screw **1050** until the bone screw portion **1060** (not visible) is secured to the glenoid **912** as previously described with respect to FIGS. **10A-10C**.

(87) FIGS. **12A**, **12B**, and **12C** provide for an exemplary embodiment of a collet removal tool **1200**. As shown the collet removal tool **1200** can include a handle portion **1210** and a distal head **1220**. For example, in some embodiments, the collet removal tool **1200** can be inserted laterally or proximally into the central throughbore **310** of the metaglene **210**. An operator can rotate or otherwise actuate the handle portion **1210** such that the distal head **1220** engages the inner threaded portion **510** of the proximally extending arms **506** of the collet **230**.

(88) In some embodiments, the distal head **1220** can include a first engagement mechanism **1232** having a threaded external surface configured to engage the thread portion **510** of the collet **230** as the head is rotated. In some embodiments, the first engagement mechanism **1232** can be a threaded cylindrical body disposed at a distal most end of the head **1220**.

(89) As previously discussed with respect to FIG. **4C**, a lip **514** of the collet **230** can be secured within a catch **324** or other recessed sidewall portion defined in the throughbore **310**. To disengage the lip **514** from the catch **324**, the distal head **1220** can include a second engagement mechanism **1234** proximal to the first engagement mechanism **1232**. As the head of the tool **1220** is rotated and advanced laterally or proximally, the second engagement mechanism **1234** can be configured to radially compress the arms **506** of the collet **230** radially inward away from the catch **324** in the throughbore.

(90) For example, in some embodiments, the second engagement mechanism **1234** can have a distal-facing bearing surface that conforms to the shape or profile at the proximal end of the collet **230**. As previously described with respect to FIG. **5**, the collet **230** can have a conical or tapered

bearing surface **516** at the proximal end of each arm **506** to facilitate removal. Thus, in some embodiments, the second engagement mechanism **1234** can be configured to have a distal-facing bearing surface of the distal head **1220** that complements the collet's conical or tapered bearing surface **516**.

(91) As shown in the illustrated embodiment of FIG. **12C**, as the operator continues to rotate the handle portion **1210**, the head of the tool **1220** can advance laterally or proximally until the distal facing feature surface of the second engagement mechanism **1234** engages the conical or tapered bearing surface **516** of the collet. As the head **1220** further advances, the second engagement mechanism **1234** can radially compress the arms of the collet away from the catch **324** or other recessed sidewall portion and thereby release the lip **514** of collet **230** from the catch. Once the lip **514** of the collet **230** is released from the catch **324**, the distal head **1220** can be pulled proximally out of the central throughbore **310** along with the collet which is threadably attached to the first engagement mechanism **1232**.

(92) Although the collet removal tool **1200** is described above for removing a collet **230** which is separate from the bone screw **270**, one of ordinary skill in the art will recognize that the collet remove tool **1200** can be used to remove an integrated collet/bone screw **1050** from with the central throughbore of a metaglene in a substantially similar manner.

(93) FIGS. **13A** and **13B** are schematic illustrations of an exemplary embodiment of a shoulder joint implant **1300** having a central bone screw integrated into the metaglene. In the illustrated embodiment, the shoulder joint implant **1300** can include a metaglene **1310** and a glenosphere **1350**. The metaglene **1310**, sometimes referred to herein as a monoblock construct, can include platform **1312** having a proximal-facing surface **1314** and a distal-facing surface **1316**. Peripheral bone screws (not shown) can be positioned in some or all of the peripheral screw holes **1318** defined in the platform **1312** and driven into the glenoid to fix the metaglene **1310** in place. In some embodiments, a bone screw portion **1320** can extend outwardly from the distal-facing surface **1316** of the platform **1312** along a central longitudinal axis of the metaglene **1310**. As described in more detail with respect to FIG. **15**, the bone screw portion **1320** can be screwed into the bony anatomy of the glenoid at the same time that the metaglene **1310** is seated onto the glenoid, and thereby reducing the number of steps needed to install the shoulder joint implant.

(94) In some embodiments, the glenosphere **1350** can include a hemispherical body **1352** having a locking screw **1354** or other coupling element that extends outward from the distal surface **1356** of the hemispherical body **1352**. Although the illustrated embodiment disclose fixing a glenosphere **1350** to the metaglene **1310**, one of ordinary skill in the art will recognize the other prosthetic components can be coupled to the metaglene, such as but not limited to a glenoid component configured to replace or augment the glenoid surface of the scapula.

(95) FIG. **14** is a schematic illustration of a cross section of the metaglene **1310** of FIGS. **13A** and **13B** having an integrated central bone screw portion **1320**. In the illustrated embodiment, the metaglene **1310** includes a central throughbore **1322** configured to extend from the proximal-facing surface **1314** of the platform **1312** and partially into the bone screw portion **1320**. The locking screw **1354** of the glenosphere **1350** can be screwed directly into the central throughbore **1322** to draw the distal-facing surface **1356** of the glenosphere body **1352** onto the proximal-facing surface **1314** of the metaglene platform **1312**. In some embodiments, a proximal end of the central throughbore **1322p** can be configured to threadably engage central locking screw **1354** of the glenosphere **1350**. As shown in the illustrated embodiment, the outer thread diameter of the bone screw portion **1320** can be greater at a proximal end portion **1320p** than the outer thread diameter at a distal end portion **1320d** in order to accommodate to the diameter of the locking screw **1354**.

(96) FIG. **15** illustrates an exemplary embodiment of a method **1500** for implanting the shoulder joint implant **1300** of FIGS. **13A** and **13B** into a patient's scapula. In the illustrated embodiment, the shoulder joint implant **1300** can be implanted as part of a reverse shoulder reconstruction surgery.

(97) At block **1510**, the metaglene **1310** having an integrated central bone screw portion, or monoblock construct, can be attached to a distal end of a monoblock delivery instrument **800**. In some embodiments, the monoblock delivery instrument **800** can have a handle portion **810**, a distal head **820**, and a locking mechanism **830**. The distal head **820** can be specifically adapted to engage one or more of the screw holes **1318** or other locking feature defined on the proximal face **1314** of the platform **1312**.

(98) At block **1520**, the monoblock construct **1310** can be locked to the monoblock delivery instrument **800** using the locking mechanism **830**. In some embodiments, the locking mechanism **830** can include a handle portion **832** and an elongated shaft **834** having a distal threaded head **834**. The locking mechanism **830** can be laterally or proximally inserted through a cannula defined along a central axis of the handle portion **810** and the distal head **820** and received within the central throughbore **1322** (not visible) of the monoblock construct **1310**. The handle portion **832** of the locking mechanism **830** can be rotated in a direction (e.g., clockwise) such that the threaded head **836** of the locking mechanism **830** threadably engages the central throughbore and thereby locks the monoblock construct **1310** to the distal head **820** of the monoblock delivery instrument **800**.

(99) At block **1530**, the monoblock construct **1310** can be inserted into a hole drilled into the glenoid **912** of a patient's scapula **914**. For example, as shown in the illustrated embodiment, the monoblock delivery instrument **800** can push the bone screw portion **1320** extending from the monoblock construct **1310** into the hole.

(100) At block **1540**, the monoblock construct **1310** can be seated onto the glenoid **912**, such that the bone screw portion **1320** (not visible) is screwed into the glenoid vault (not visible) and the distal-facing surface **1316** (not visible) of the metaglene platform **1312** is seated on the bony surface of the glenoid **912**. As shown in the illustrated embodiment, the monoblock delivery instrument **800** can be rotated in a tightening direction (e.g., clockwise) to screw the bone screw portion **1320** until a distal-facing surface **1316** (not visible) of the monoblock construct **1310** is seated onto the glenoid.

(101) At block **1550**, the peripheral screw holes **1318** of the platform **1312** of the monoblock construct **1310** can be aligned. In some embodiments, the handle portion **810** of the monoblock delivery instrument **800** can be rotated as needed in a clockwise and/or a counter clockwise direction to align the peripheral screw holes **1318** such that peripheral screws (not shown) can be inserted and driven through the screw holes in one or more of a superior, inferior, medial and/or lateral position with respect to the surface of the glenoid bone.

(102) At block **1560**, the handle portion **832** of the locking mechanism **830** can be rotated in a direction to unlock the monoblock construct **1310** from the distal head **820** of the monoblock delivery instrument **800**. For example, as shown in the illustrated embodiment, the handle portion **832** of the locking mechanism **830** can be rotated counter-clockwise to disengage the threaded head **836** (not visible) of the locking mechanism **830** from the central throughbore (e.g., **1322**, not visible) of the monoblock construct **1310** and thereby unlock the monoblock construct **1310** from the distal head **820** of the monoblock delivery instrument **800**.

(103) At block **1570**, the monoblock delivery instrument **800** can be removed from the monoblock construct **1310** by proximally pulling the instrument away from the construct. After the monoblock inserter instrument is removed, peripheral screws can be inserted and driven through the peripheral screw holes **1318** in the platform **1312** of the monoblock construct **1310**. Alternatively or additionally, the glenosphere **1350** (not shown) can be secured to the platform **1312** of the monoblock construct **1310** by screwing the locking screw **1354** (not shown) into the threaded portion of the central throughbore **1322** (not shown).

(104) The preceding description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the claims. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied

to other embodiments without departing from the scope of the claims. Thus, the present disclosure is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the following claims and the principles and novel features disclosed herein. All publications and references cited herein are expressly incorporated herein by reference in their entirety.

Claims

1. A collet for use in a shoulder joint implant, comprising: a cylindrical body having an annular base portion on a distal portion thereof, at least three radially compressible arms, and a plurality of open-ended slots disposed between the radially compressible arms, each slot of the plurality of open-ended slots having a base end that is closed and terminates proximate to or at the annular base portion and an open-end that is open and terminates at a proximal-most end of the cylindrical body, the open-ended slots being spaced about a circumference of the cylindrical body; one or more lateral protrusions protruding radially outward from at least one of a radially compressible arm of the at least three radially compressible arms or the annular base portion, the one or more lateral protrusions being configured to provide resistance to torsional rotation; and an annular protrusion disposed on a proximal portion of the cylindrical body, distal of the proximal-most end of the cylindrical body, the annular protrusion protruding radially outward from a main body portion of one or more radially compressible arms of the at least three radially compressible arms, and the annular protrusion being configured to secure the collet at a predetermined depth.
2. The collet of claim 1, wherein the at least three radially compressible arms are configured to flex in response to changes in a diameter of a throughbore of a baseplate when the collet is passed into the throughbore.
3. The collet of claim 1, wherein the at least three radially compressible arms comprises four radially compressible arms.
4. The collet of claim 3, wherein each of the four radially compressible arms comprises an inner threaded portion configured to at least partially surround and threadably engage a locking screw associated with a prosthetic component such that the prosthetic component is drawn onto a proximal-facing bearing surface of a baseplate to which the prosthetic component is mated.
5. The collet of claim 3, wherein the one or more lateral protrusions protruding radially outward from at least one of a radially compressible arm of the at least three radially compressible arms or the annular base portion comprises a proximal lateral protrusion protruding radially outward from each of the four radially compressible arms and four distal lateral protrusions protruding radially outward from the annular base portion, the four distal lateral protrusions being vertically aligned with each of the proximal lateral protrusions.
6. The collet of claim 1, wherein the one or more lateral protrusions protruding radially outward from at least one of a radially compressible arm of the at least three radially compressible arms or the annular base portion comprises a proximal lateral protrusion protruding radially outward from each of the at least three radially compressible arms and distal lateral protrusions protruding radially outward from the annular base portion, the distal lateral protrusions being vertically aligned with each of the proximal lateral protrusions.
7. The collet of claim 1, wherein each of the at least three radially compressible arms comprises an inner threaded portion configured to at least partially surround and threadably engage a locking screw associated with a prosthetic component such that the prosthetic component is drawn onto a proximal-facing bearing surface of a baseplate to which the prosthetic component is mated.
8. The collet of claim 1, wherein the one or more lateral protrusions protruding radially outward from at least one of a radially compressible arm of the at least three radially compressible arms or the annular base portion comprises a first lateral protrusion protruding radially outward from the radially compressible arm of the at least three radially compressible arms and a second lateral

protrusion protruding radially outward from the annular base portion.

9. The collet of claim 1, wherein each lateral protrusion of the one or more lateral protrusions are configured to slide along a complementary keyway of a plurality of keyways formed along an inner sidewall of a throughbore of a baseplate.

10. The collet of claim 1, wherein the annular protrusion is configured to engage a catch defined within a throughbore of a baseplate to secure the collet in place with respect to the baseplate at the predetermined depth.

11. The collet of claim 1, further comprising: a tapered proximal bearing surface disposed proximal of the annular protrusion and configured to facilitate removal of the collet from within a throughbore of a baseplate.

12. The collet of claim 1, wherein a diameter of the cylindrical body is approximately in the range of about 3 millimeters to about 15 millimeters, and wherein a height of the cylindrical body is approximately in a range of about 5 millimeters to about 20 millimeters.

13. A shoulder joint implant, comprising: a prosthetic component that includes a coupling element protruding from a distal bearing surface thereof; a baseplate configured to secure the prosthetic component to bone, the baseplate including a post protruding from a distal bearing surface thereof and defining a throughbore that extends along a proximal-distal axis of the post; and the collet of claim 1 configured to be disposed within the throughbore of the baseplate such that the at least three radially compressible arms engage the coupling element of the first prosthetic component to fixedly couple to the baseplate.

14. The shoulder joint implant of claim 13, wherein the coupling element comprises a locking screw and that engagement of the locking screw by the at least three radially compressible arms causes the prosthetic component to be drawn onto a proximal-facing bearing surface of the baseplate.

15. The shoulder joint implant of claim 13, wherein the at least three radially compressible arms are configured to flex in response to changes in a diameter of the throughbore of the baseplate when the collet is passed into the throughbore.

16. The shoulder joint implant of claim 13, wherein the at least three radially compressible arms comprises four radially compressible arms.

17. The shoulder joint implant of claim 16, wherein each of the four radially compressible arms comprises an inner threaded portion configured to at least partially surround and threadably engage the coupling element of the prosthetic component such that the prosthetic component is drawn onto a proximal-facing bearing surface of the baseplate.

18. The shoulder joint implant of claim 13, wherein the one or more lateral protrusions protruding radially outward from at least one of a radially compressible arm of the at least three radially compressible arms or the annular base portion comprises a proximal lateral protrusion protruding radially outward from each of the at least three radially compressible arms and distal lateral protrusions protruding radially outward from the annular base portion, the distal lateral protrusions being vertically aligned with each of the proximal lateral protrusions.

19. The shoulder joint implant of claim 13, wherein each lateral protrusion of the one or more lateral protrusions are configured to slide along a complementary keyway of a plurality of keyways formed along an inner sidewall of the throughbore of the baseplate.

20. The shoulder joint implant of claim 13, wherein the annular protrusion is configured to engage a catch defined within the throughbore of the baseplate to secure the collet in place with respect to the baseplate at the predetermined depth.
